



RGPV Bhopal

Information Technology

ACADEMIC PERFORMANCE REPORT

AI-Powered Comprehensive Analytics Report

PROGRAM

B.Tech.

SEMESTER

Semester 3

GENERATED

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Note: Page numbers are approximate and may vary based on data volume. This report contains AI-generated analytics and automatically computed insights.

EXECUTIVE SUMMARY

This report presents a comprehensive analysis of academic performance for 2 students enrolled in the **B.Tech.** program, Semester 3 at **RGPV Bhopal**. The analysis covers overall statistics, branch-wise comparisons, subject-level insights, grade distributions, risk assessments, and AI-generated recommendations for academic improvement.

KEY PERFORMANCE INDICATORS

2 Total Students	100.0% Pass Rate	0.0% Fail Rate	8.13 Avg SGPA
8.18 Avg CGPA	8.13 Highest SGPA	8.13 Lowest SGPA	8.13 Median SGPA

KEY FINDINGS

- Excellent batch performance** — 100.0% of students passed all subjects, indicating strong academic standards.
- Batch Topper:** AASTHA AGRAWAL (0105IT241001) achieved the highest SGPA of 8.13.
- The average SGPA of **8.13** reflects strong overall academic performance.
- Best performing department:** Information Technology.

PERFORMANCE SNAPSHOT

Metric	Value
Total Students Evaluated	2
Students Passed	1 (100.0%)
Students Failed	0 (0.0%)
Average SGPA	8.13
Median SGPA	8.13
Standard Deviation	0.0

Metric	Value
Average CGPA	8.18
Highest SGPA	8.13 (AASTHA AGRAWAL)
Lowest SGPA	8.13
Departments Analyzed	1
Subjects Analyzed	12

OVERALL STATISTICS

Complete statistical overview of 2 students across 1 department(s) and 12 subject(s).

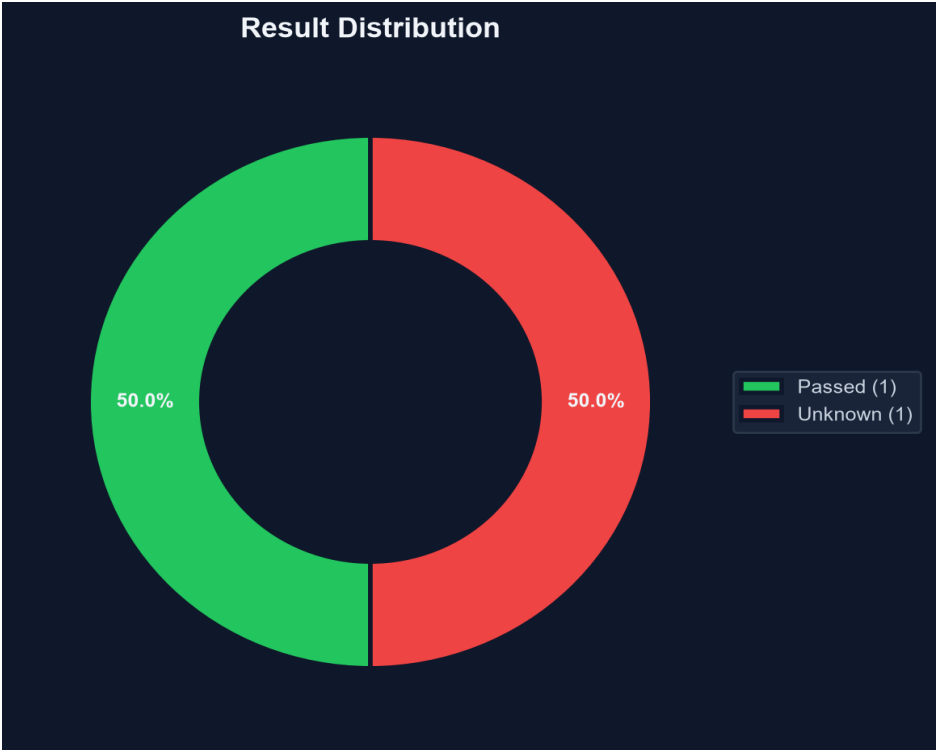
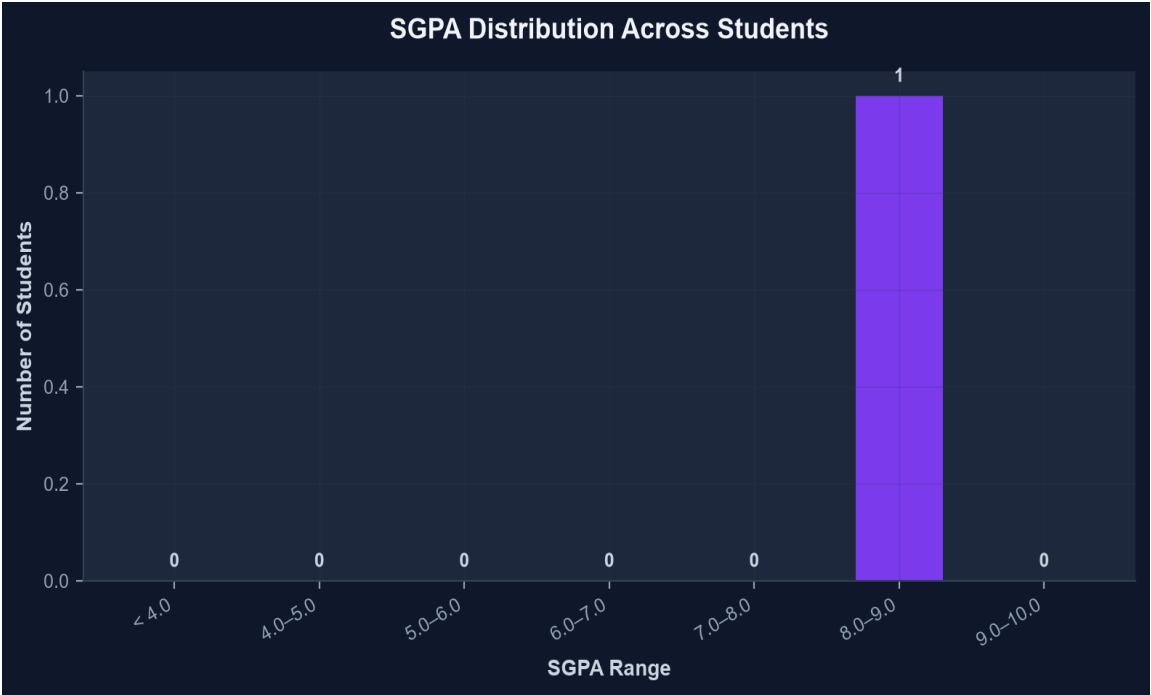
RESULT BREAKDOWN

Category	Count	Percentage
Total Students	2	100%
Passed	1	100.0%
Failed	0	0.0%
Unknown/Pending	1	50.0%

SGPA STATISTICS

Statistic	SGPA	CGPA
Highest	8.13	8.18
Lowest	8.13	8.18
Average (Mean)	8.13	8.18
Median	8.13	8.18
Std. Deviation	0.0	—

SGPA DISTRIBUTION



BRANCH-WISE ANALYSIS

Comparative performance analysis across 1 academic department(s). This section evaluates each branch based on average SGPA, pass rate, failure rate, and backlog distribution.

DEPARTMENT COMPARISON TABLE

Branch	Students	Pass %	Fail %	Avg SGPA	Avg CGPA	Backlogs	Topper
IT	2	100.0%	0.0%	8.13	8.18	0	AASTHA AGRAWAL

SEMESTER ANALYSIS

Performance analysis for **Semester 3** of the **B.Tech.** program. This section provides a snapshot of the semester's academic outcomes.

SEMESTER PERFORMANCE OVERVIEW

Metric	Value
Semester	Semester 3
Program	B.Tech.
Total Students	2
Pass Count	1
Fail Count	0
Pass Percentage	100.0%
Average SGPA	8.13
Average CGPA	8.18
Highest SGPA	8.13
Lowest SGPA	8.13
Number of Subjects	12
Departments	1

DEPARTMENT-WISE SEMESTER RESULTS

Department	Students	Pass %	Avg SGPA	Topper
Information Technology	2	100.0%	8.13	AASTHA AGRAWAL

Note: Multi-semester trend analysis requires comparison data from previous semesters. Use the --compare flag with a previous semester's Excel file to enable trend charts.

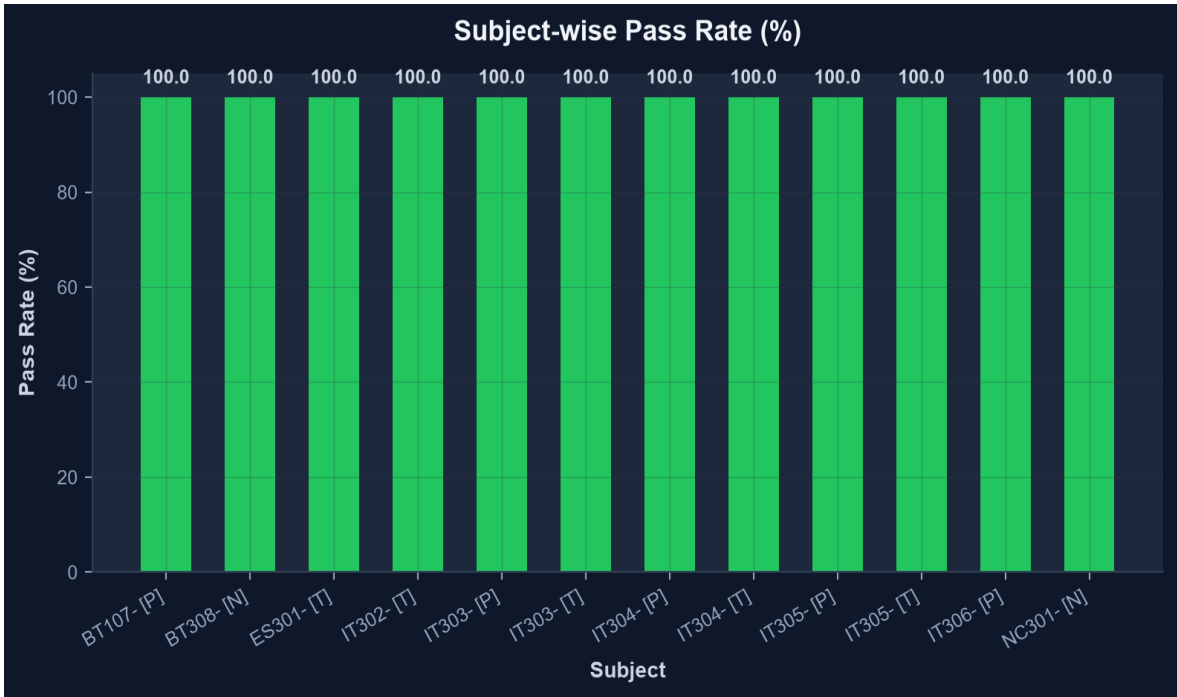
SUBJECT-WISE ANALYSIS

Detailed analysis of **12** subject(s) evaluated in this semester. Each subject is assessed for pass rate, failure rate, average grade, difficulty index, and grade distribution.

SUBJECT PERFORMANCE SUMMARY

Subject	Students	Pass %	Fail %	Avg GP	Difficulty	Highest	Lowest
BT107- [P]	1	100.0%	0.0%	8.0	0.2	A	A
BT308- [N]	1	100.0%	0.0%	0.0	1.0	—	—
ES301- [T]	1	100.0%	0.0%	7.0	0.3	B+	B+
IT302- [T]	1	100.0%	0.0%	7.0	0.3	B+	B+
IT303- [P]	1	100.0%	0.0%	9.0	0.1	A+	A+
IT303- [T]	1	100.0%	0.0%	6.0	0.4	B	B
IT304- [P]	1	100.0%	0.0%	8.0	0.2	A	A
IT304- [T]	1	100.0%	0.0%	7.0	0.3	B+	B+
IT305- [P]	1	100.0%	0.0%	9.0	0.1	A+	A+
IT305- [T]	1	100.0%	0.0%	6.0	0.4	B	B
IT306- [P]	1	100.0%	0.0%	8.0	0.2	A	A
NC301- [N]	1	100.0%	0.0%	0.0	1.0	—	—

SUBJECT-WISE PASS RATE



SUBJECT DIFFICULTY INDEX

The Difficulty Index ranges from 0.0 (easiest) to 1.0 (hardest), calculated based on average grade points. Higher values indicate more challenging subjects.



DETAILED SUBJECT BREAKDOWN

- **BT107- [P]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 1]
- **BT308- [N]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 1]
- **ES301- [T]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 7.0, Failed: 0 — Grades: [B+: 1]
- **IT302- [T]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 7.0, Failed: 0 — Grades: [B+: 1]
- **IT303- [P]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 9.0, Failed: 0 — Grades: [A+: 1]
- **IT303- [T]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 6.0, Failed: 0 — Grades: [B: 1]
- **IT304- [P]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 1]
- **IT304- [T]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 7.0, Failed: 0 — Grades: [B+: 1]
- **IT305- [P]:** 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 9.0, Failed: 0 — Grades: [A+: 1]

- **IT305- [T]**: 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 6.0, Failed: 0 — Grades: [B: 1]
- **IT306- [P]**: 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 1]
- **NC301- [N]**: 1 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 1]

✓ **Easiest Subject: IT303- [P] (Difficulty Index: 0.1)**

TOPPER ANALYSIS

Academic rank list of top-performing students. Rankings are based on Semester GPA (SGPA), with CGPA as a tiebreaker.

TOP 10 STUDENTS

Rank	Enrollment	Name	Branch	SGPA	CGPA	Result
■	0105IT241001	AASTHA AGRAWAL	IT	8.13	8.18	PASS

TOP STUDENTS — SGPA VISUALIZATION



STUDENT PERFORMANCE DETAILS

Individual performance cards for 2 student(s). Each card shows subject grades, SGPA, CGPA, result status, and backlog count.

#1 AASTHA AGRAWAL0105IT241001

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B+
IT302- [T]	B+
IT303- [T]	B
IT304- [T]	B+
IT305- [T]	B
BT107- [P]	A
IT303- [P]	A+
IT304- [P]	A
IT305- [P]	A+
IT306- [P]	A
BT308- [N]	NA
NC301- [N]	NA

SGPA

8.13

CGPA

8.18

RESULT

PASS

BACKLOGS

0

#2 0105IT2410020105IT241002

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
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SGPA

CGPA

RESULT

UNKNOWN

BACKLOGS

0

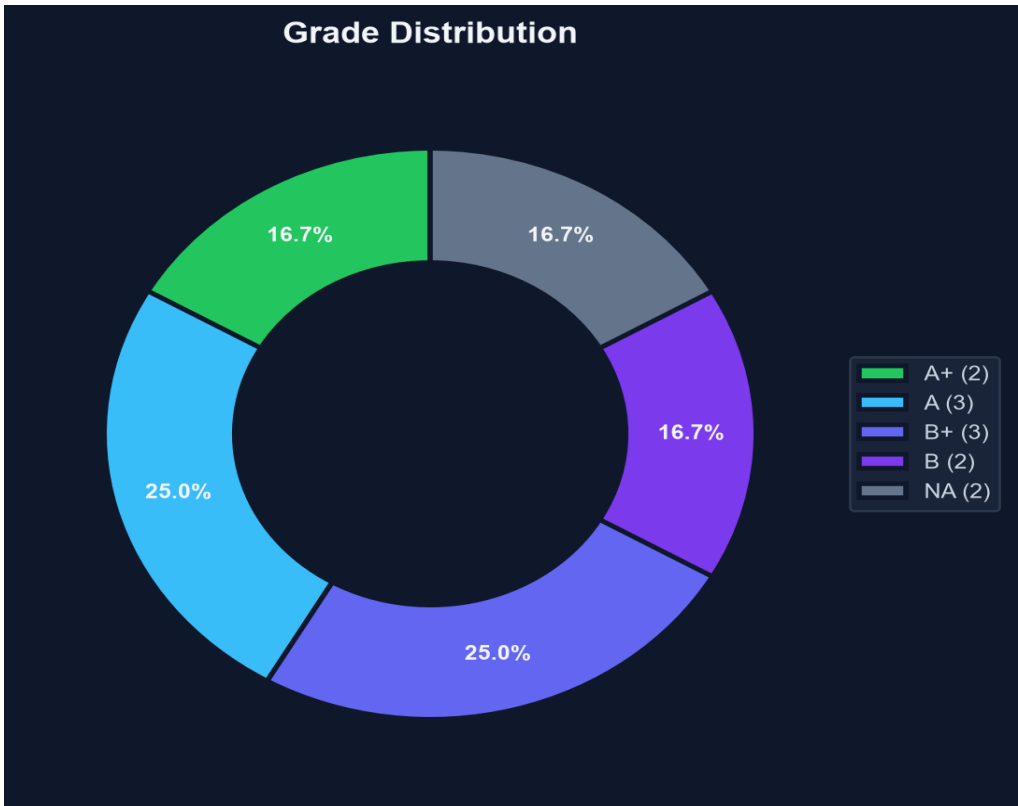
GRADE DISTRIBUTION

Distribution analysis of 12 total grades awarded across all subjects and students.

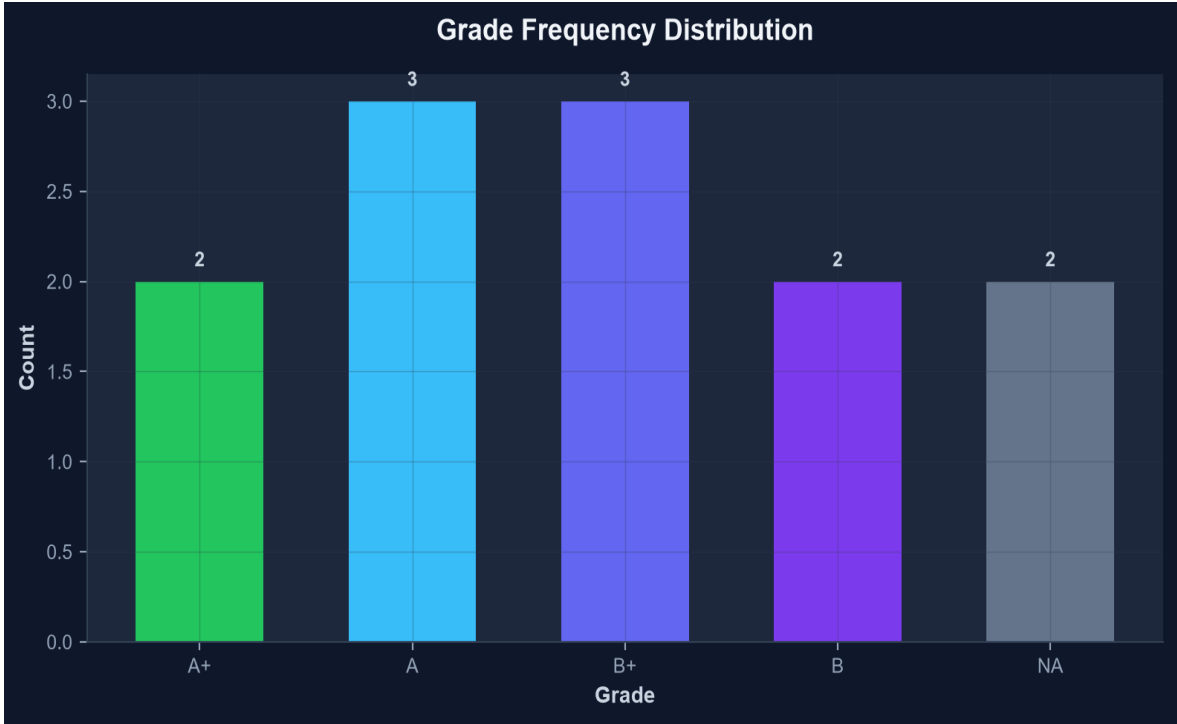
GRADE FREQUENCY TABLE

Grade	Count	Percentage	Performance Level
A+	2	16.7%	Excellent
A	3	25.0%	Very Good
B+	3	25.0%	Good
B	2	16.7%	Above Average
NA	2	16.7%	NA

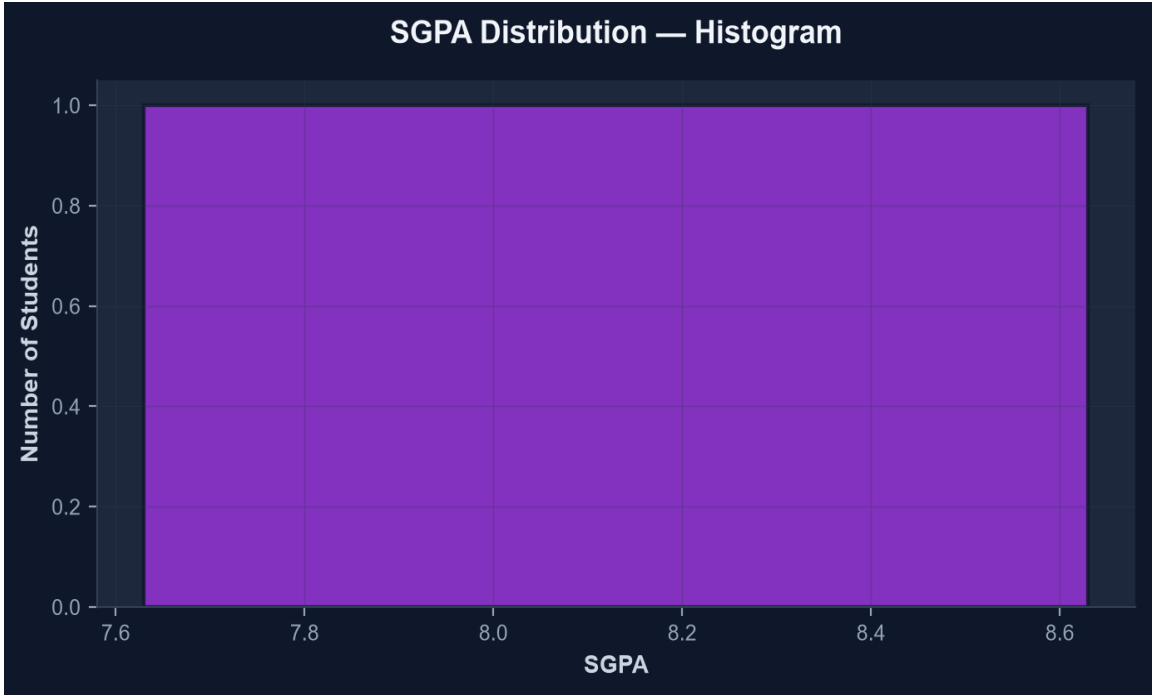
GRADE DISTRIBUTION — DONUT CHART



GRADE DISTRIBUTION — BAR CHART



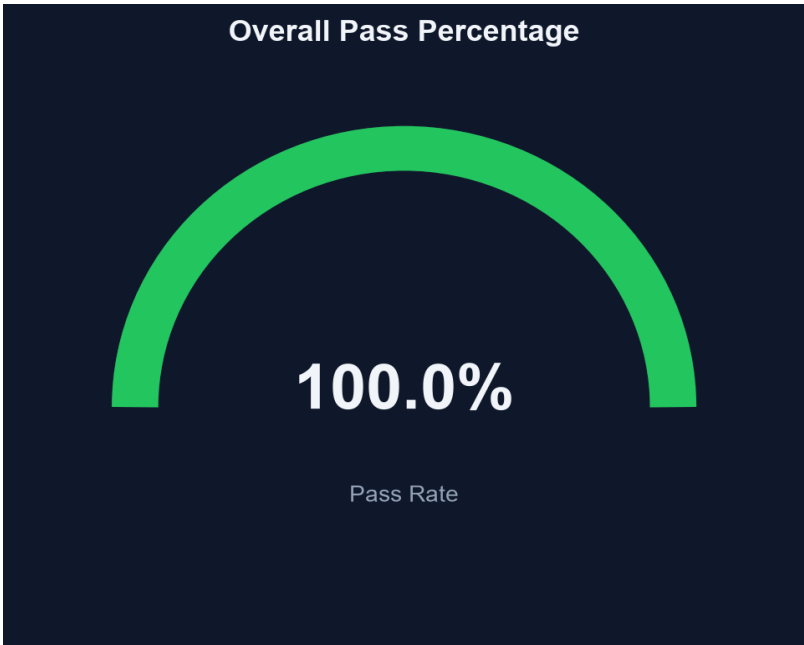
SGPA FREQUENCY HISTOGRAM



PASS PERCENTAGE ANALYSIS

Overall pass rate analysis showing the proportion of students who cleared all subjects versus those with one or more failures.

OVERALL PASS RATE GAUGE



DEPARTMENT-WISE PASS RATE

Department	Total	Passed	Failed	Pass %
Information Technology	2	1	0	100.0%



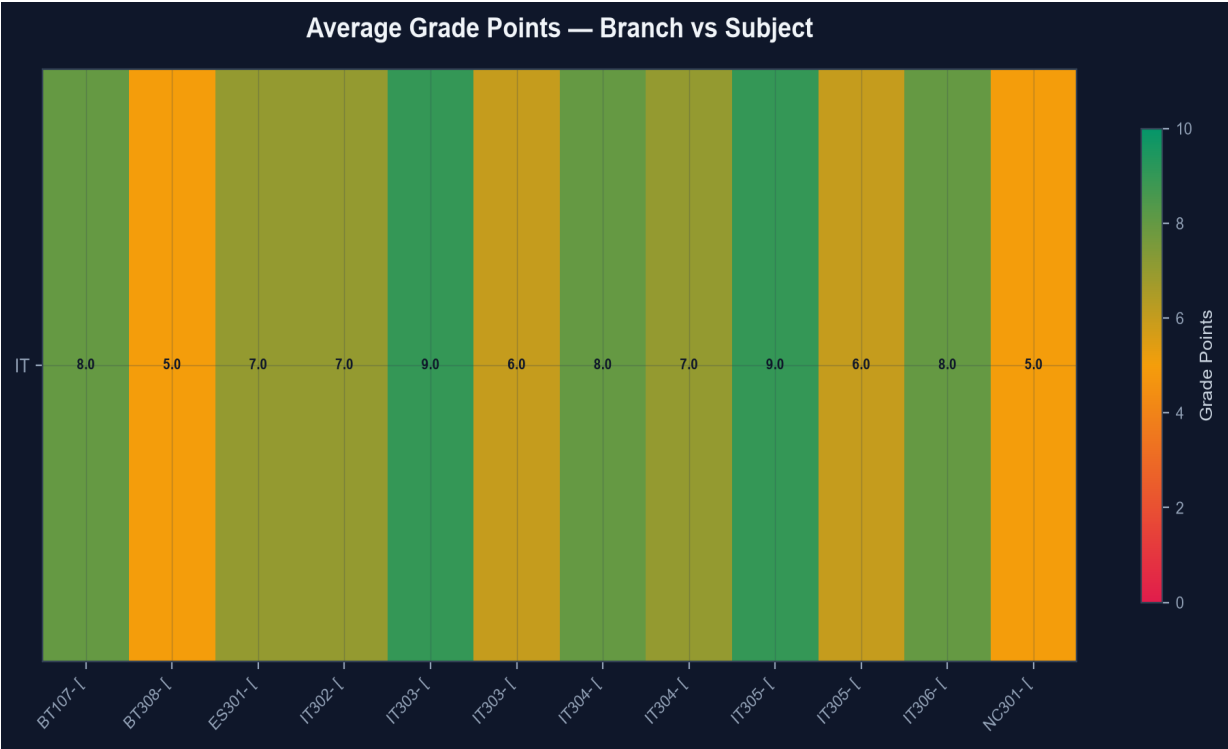
BACKLOG ANALYSIS

Analysis of **0** total backlog instances across **0** student(s). Backlogs include F, Fail, AB, Absent, and RL grades.

PERFORMANCE HEATMAPS

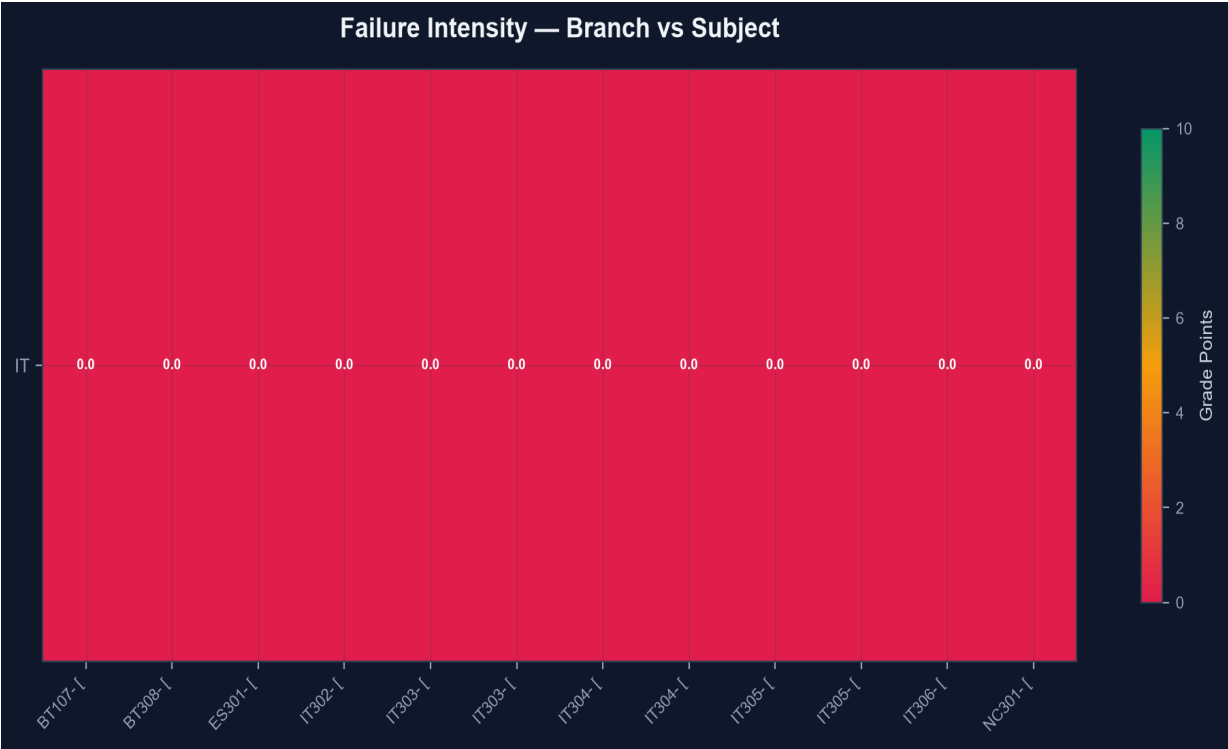
Visual heatmaps showing performance intensity across branches and subjects. Warmer colors (red/orange) indicate weaker performance; cooler colors (green) indicate stronger results. Grade points range from 0 (Fail) to 10 (Outstanding).

BRANCH × SUBJECT PERFORMANCE HEATMAP



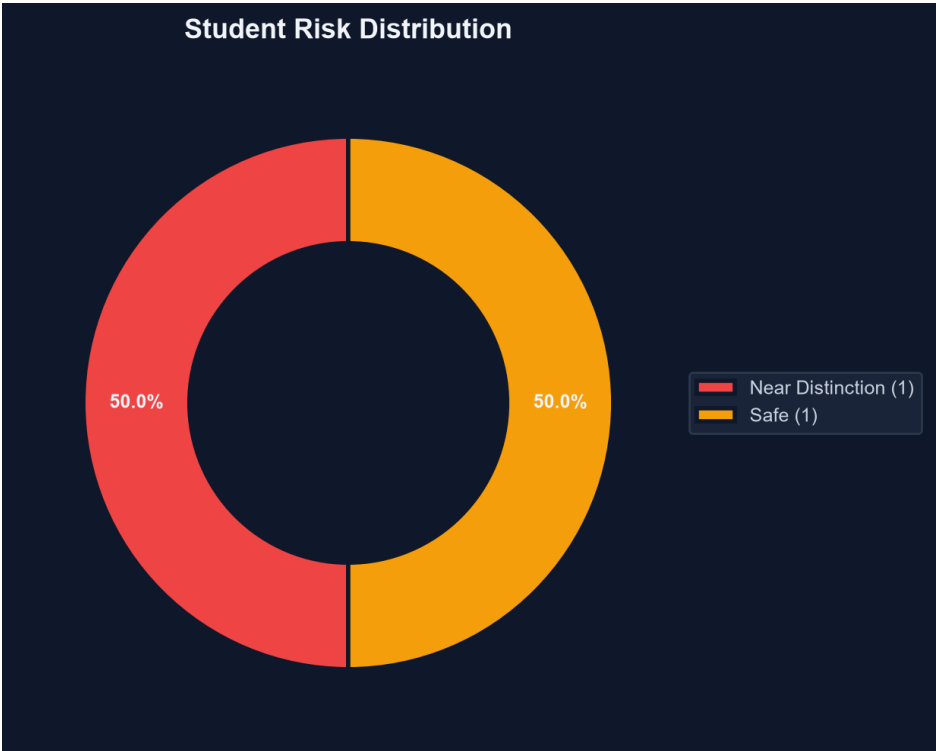
FAILURE INTENSITY HEATMAP

Higher values (red) indicate higher failure rates. Scale: 0 = No Failures, 10 = 100% Failure.



RISK ANALYSIS

Academic risk stratification identifying students who need attention. **0** at-risk, **1** near-distinction, **0** with multiple backlogs.



STUDENTS NEAR DISTINCTION (SGPA 7.5–8.5)

Enrollment	Name	Branch	SGPA	CGPA
0105IT241001	AASTHA AGRAWAL	Information Tec	8.13	8.18

IMPROVEMENT ANALYSIS

Improvement analysis requires comparison data from a previous semester or job. To enable this section, generate the report with the **--compare** flag pointing to a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

COMPARISON ANALYSIS

Comparison analysis is available when a previous dataset is provided for reference. Use the **--compare** flag to specify a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

AI-GENERATED INSIGHTS

The AI analytics engine has generated **26** observations based on comprehensive statistical analysis of the academic data. These insights are automatically derived from patterns, distributions, and trends.

- 1 A total of 2 students were analyzed in this academic evaluation.
- 2 Overall pass percentage stands at 100.0%, with 1 students clearing all subjects.
- 3 Outstanding batch performance — over 90% pass rate indicates strong academic preparedness.
- 4 Average SGPA across all students is 8.13 out of 10.0.
- 5 Median SGPA is 8.13, indicating the central tendency of student performance.
- 6 Highest SGPA achieved is 8.13 by AASTHA AGRAWAL (0105IT241001).
- 7 Lowest SGPA recorded is 8.13.
- 8 Average Cumulative GPA (CGPA) is 8.18.
- 9 Information Technology is the best-performing department based on average SGPA.
- 10 Information Technology is the weakest-performing department and may need academic support.
- 11 Information Technology department: 2 students, 100.0% pass rate, average SGPA 8.13.
- 12 [TOP] Information Technology achieved a perfect 100% pass rate — exemplary performance.
- 13 Subject BT308- [N] has a high difficulty index of 1.0 (scale 0-1).
- 14 Subject NC301- [N] has a high difficulty index of 1.0 (scale 0-1).
- 15 1 student(s) are near distinction level (SGPA 7.5–8.5) — targeted support could push them higher.
- 16 'A+' grade was the achieved 2 times (16.7% of all grades).
- 17 1 students (50.0%) scored SGPA above 8.0, demonstrating strong academic excellence.

-
- 18 Batch topper is AASTHA AGRAWAL (0105IT241001) with SGPA 8.13.
-
- 19 This report covers a cohort of 2 students from the AASTHA batch.
-
- 20 Out of 2 students evaluated, 1 successfully cleared all subjects.
-

-
- 21 Academic performance data has been analyzed across multiple dimensions including branch, subject, and individual student levels.
 - 22 Statistical measures including mean, median, and standard deviation have been computed for comprehensive analysis.
 - 23 Grade distribution analysis reveals the spread of academic achievement across all evaluated subjects.
 - 24 Backlog analysis identifies subjects and departments requiring targeted academic interventions.
 - 25 Risk stratification categorizes students based on their likelihood of academic difficulty.
 - 26 This AI-generated report provides data-driven insights for informed academic decision-making.
-

AI RECOMMENDATIONS

The AI engine has generated 15 actionable recommendations based on the analysis findings. Recommendations are prioritized as **HIGH**, **MEDIUM**, and **LOW** priority.

Priority Distribution: 0 **HIGH** | 7 **MEDIUM** | 8 **LOW**

MEDIUM	R1	Review curriculum difficulty for BT308- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R2	Review curriculum difficulty for NC301- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R3	Provide targeted coaching to 1 near-distinction student(s) — small improvement could push them to distinction level.
MEDIUM	R4	Increase practical and hands-on laboratory sessions to improve conceptual understanding.
MEDIUM	R5	Implement continuous internal assessment to provide early warning signals for academic difficulties.
MEDIUM	R6	Organize industry expert guest lectures to improve student engagement and practical understanding.
MEDIUM	R7	Implement a digital learning management system for supplementary resources and self-paced revision.
LOW	R8	Recognize and reward top-performing students with merit certificates and academic incentives.
LOW	R9	Encourage toppers to participate in peer tutoring programs to uplift overall batch performance.
LOW	R10	Conduct student feedback surveys on teaching quality and curriculum relevance.
LOW	R11	Establish study groups and collaborative learning environments for improved peer learning.
LOW	R12	Review and update course materials to align with current industry standards and practices.

- LOW

R13

Organize academic orientation sessions at the start of each semester to set performance expectations.
- LOW

R14

Publish this analytics report to department heads and academic coordinators for data-driven planning.
- LOW

R15

Schedule regular parent-teacher meetings for students with declining performance trends.